# Chap 4 Metal-Insulator-semiconductor capacitors

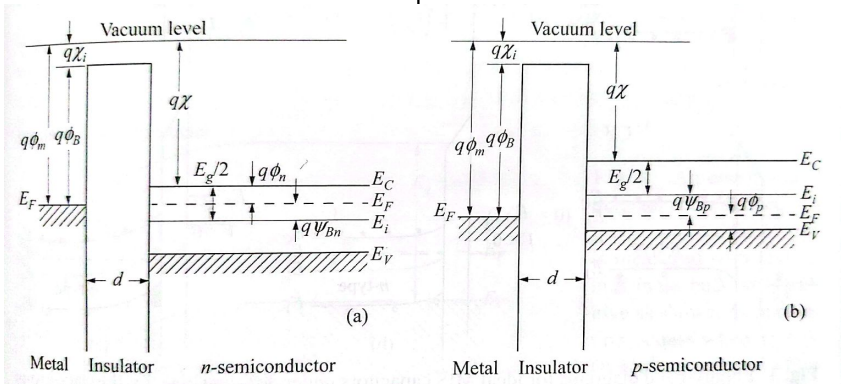



**Fig. 2** Energy-band diagrams of ideal MIS capacitors at equilibrium ($V = 0$). (a) An $n$-type semiconductor. (b) A $p$-type semiconductor.

Ideal MIS capacitor:

1) Only charges:

$Q_M = -Q_S$ ← In semiconductor.

↑ In metal

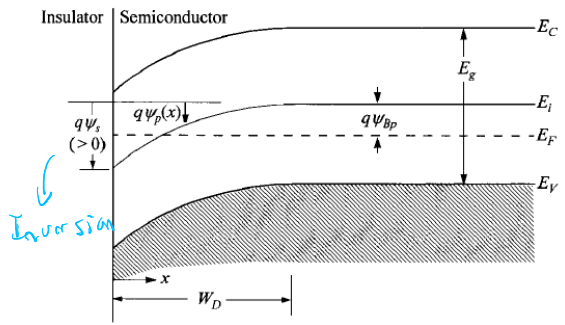
2) $R_I \to \infty \to I = 0 \to E_F$ cte

↓ $\neq \rho(x)$

3) $\phi_M - \phi_S = \phi_{MS} = 0$

For p semiconductor:

$$q\psi_B = q\psi_{Bp} = E_i(x \to \infty) - E_F \quad (4.4)$$




→ $q\psi_p = E_i(\infty) - E_i(x)$ (4.4)

↑ Bands bend

→ $\beta = \frac{q}{kT}$ (4.3)

→ $\psi_s = \psi_p(x=0)$

↑ surface potential

→ $p_{p_0} = n_i e^{(E_i - E_F)/kT} = n_i e^{\beta\psi_{B_p}} = N_A$

$x \to \infty$

→ $p_p(x) = n_i e^{[E_i(x) - E_F]/kT}$

$q\psi_{Bp} - q\psi_p(x)$

$p_p(x) = p_{p_0} e^{-\beta\psi_p(x)}$

→ $n_{p_0} = n_i e^{[E_F - E_i(\infty)]/kT} = n_i e^{-\beta\psi_{B_p}}$

→ $n_p(x) = n_i e^{[E_F - E_i(x)]/kT}$

$q\psi_p(x) - q\psi_{Bp}$

$n_p(x) = n_i e^{\beta\psi_p(x)} e^{-\beta\psi_{B_p}}$

$n_p(x) = n_{p_0} e^{\beta\psi_p(x)}$

At the interface of I-S:

$$p_p(x=0) = p_{p_0} e^{-\beta\psi_s} \quad (4.6)$$

($p_{p_0} \nearrow N_A$)

$$n_p(x=0) = n_{p_0} e^{\beta\psi_s}$$

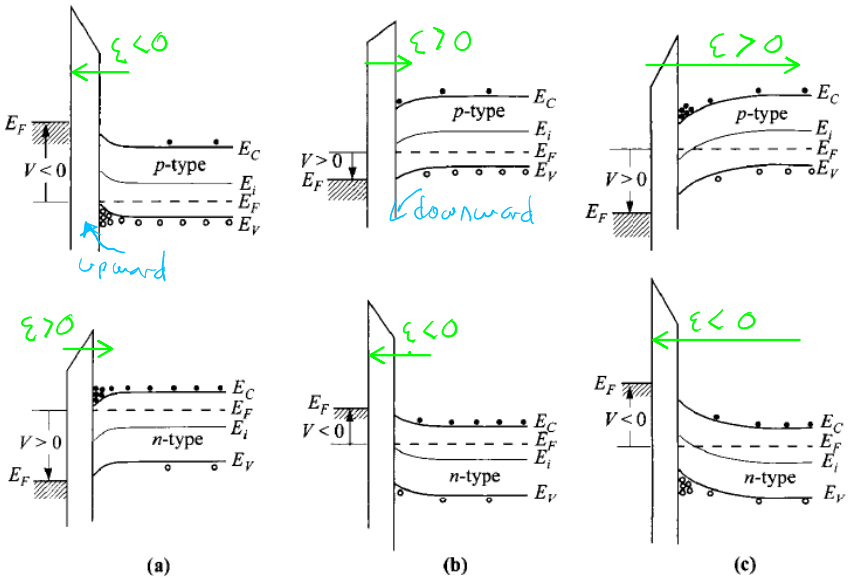



**Fig. 3** Energy-band diagrams for ideal MIS capacitors under different bias, for the conditions of: (a) accumulation, (b) depletion, and (c) inversion. Top/bottom figures are for *p*-type/*n*-type semiconductor substrates.

2) Accumulation of holes:
$\psi_s < 0 \rightarrow p_p(0) \gg n_p(0)$

3) Flat-band condition:
$\psi_s = 0 \rightarrow p_p(0) \simeq p_{p_0} = N_A > n_p(0)$

3) Depletion of holes:
$0 < \psi_s < \psi_{B_p} \rightarrow N_A > p_p(0) > n_p(0)$

4) Fermi level at midgap:
$\rightarrow E_i(0) = E_F \rightarrow n_p(0) = p_p(0) = n_i$
$\rightarrow \psi_s = \psi_{B_p}$

5) Weak inversion:
$\rightarrow n_p(0) > p_p(0)$
$\rightarrow \psi_{B_p} < \psi_s < 2\psi_{B_p}$

6) Strong inversion:
$\rightarrow n_p(0) \gg p_p(0)$
$\rightarrow 2\psi_{B_p} < \psi_s$

9-1) Poisson equation inside the depletion region ↑ inside de semic.

$$\frac{d^2\psi_p}{dx^2} = -\frac{q}{\varepsilon_s}\left(p_{p0} - n_{p0} + n_p - p_p\right)$$

$$= -\frac{q}{\varepsilon_s}\left[p_{p0}\left(e^{-\beta\psi_p} - 1\right) - n_{p0}\left(e^{\beta\psi_p} - 1\right)\right] \quad (4.10)$$

$$\rightarrow n_{p0} = \frac{n_i^2}{p_{p0}}$$

Extrinsic Debye length for holes: (17)

$$L_D = \sqrt{\frac{kT\,\varepsilon_s}{p_{p0}\,q^2}} = \sqrt{\frac{\varepsilon_s}{q\,p_{p0}\,\beta}}$$

Electric field inside the depletion region:

$$\rightarrow \mathcal{E}(x) - \underbrace{\mathcal{E}(\infty)}_{0} = -\frac{d\psi_p(x)}{dx} \quad (4.19)$$

$$\mathcal{E}(x) = \begin{matrix}\psi_p > 0 \\ \pm \\ \psi_p < 0\end{matrix} \frac{\sqrt{2}}{\beta L_D} F\left(\beta\psi_p, \frac{n_{p0}}{p_{p0}}\right)$$

Definition of $F$: $\frac{n_{p0}}{p_{p0}} = e^{-2\beta\psi_{Bp}}$

$$F\left(\beta\psi_p, \frac{n_{p0}}{p_{p0}}\right) = \quad (4.18)$$

$$\sqrt{\left(e^{-\beta\psi_p} + \beta\psi_p - 1\right) + \frac{n_{p0}}{p_{p0}}\left(e^{\beta\psi_p} - \beta\psi_p - 1\right)} \geq 0$$

At the surface: $\psi_s = \psi_p(x=0)$

$$\mathcal{E}_s = \mathcal{E}(x=0) = \pm\frac{\sqrt{2}}{\beta L_D} F\left(\beta\psi_s, \frac{n_{p0}}{p_{p0}}\right) \quad (4.20)$$

Gauss law → Total space charge per unit area: (uncompensated charge)

$$Q(x=0) - \underbrace{Q(\infty)}_{\phi} = Q_s = -\varepsilon_s\mathcal{E}_s \quad (4.21)$$

$$= \begin{matrix}\psi_s > 0 \\ \pm \\ \psi_s < 0\end{matrix} \frac{\sqrt{2}\,\varepsilon_s}{\beta L_D} F\left(\beta\psi_s, \frac{n_{p0}}{p_{p0}}\right)$$

Strong inversion begins at:

$$\psi_s = 2\psi_{Bp} = \frac{2}{\beta}\ln\left(\frac{N_A}{n_i}\right) \quad (4.22)$$

Charge neutrality of the system:

$$Q_M = -Q_s = -\left(Q_n + qN_A W_D\right) \quad (4.23)$$

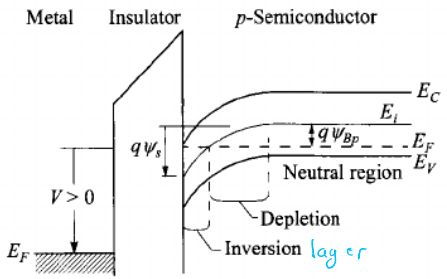
$Q_M$: metal; $Q_s$: semicond.; $Q_n$: electrons near the surface (F.4.6.b)

From Gauss law: (F.4.6)

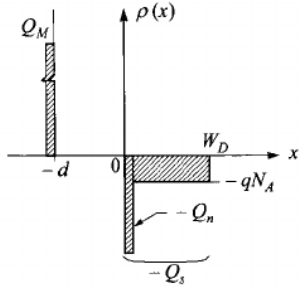
$|Q_i| = |Q_s|$ (i: insulator)

$\varepsilon_s\mathcal{E}_s = \varepsilon_i\mathcal{E}_i$

$\varepsilon_s \sim \left(\frac{3.9}{11.9}\right)\varepsilon_i$




(a)

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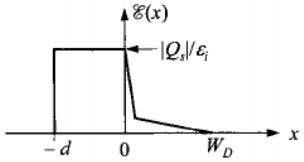
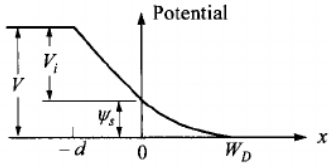



(c)




(d)

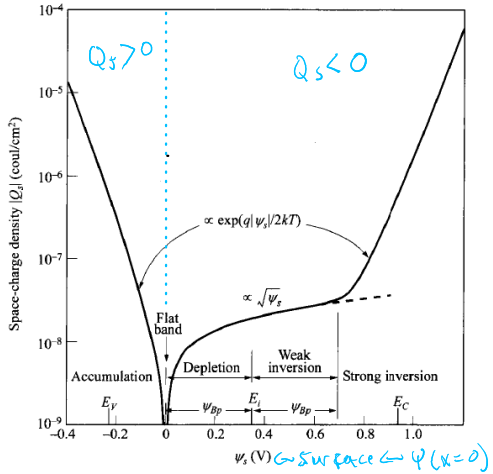


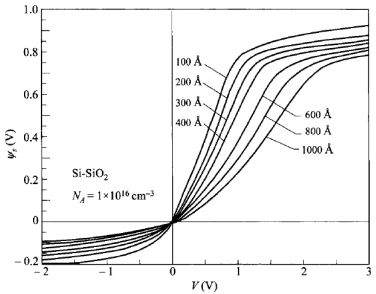

**Fig. 5** Variation of space-charge density in the semiconductor as a function of the surface potential $\psi_s$, for a $p$-type silicon with $N_A = 4\times10^{15}$ cm$^{-3}$ at room temperature.

(4.27)

Applied voltage:

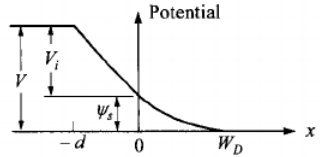
$$V = V_i + \psi_s \quad (4.24)$$

$V_i$: across the insulator; $\psi_s$: across the semiconductor




$$V_i = \varepsilon_i d = \frac{|Q_s| d}{\varepsilon_i} = \frac{|Q_s|}{C_i} \quad (4.25)$$

$C_i$: capacitance/area by electric field of insulator




$$C_i = \frac{\varepsilon_i}{d} \quad (4.26)$$

Total capacitance: (4.27)

$$C = \frac{C_i C_D}{C_i + C_D}$$

$C_D$ → depletion-layer capacitance

Depletion-layer capacitance:

$$C_D = \frac{dQ_s}{d\psi_s} \quad (4.28)$$

$$= \frac{\varepsilon_s}{\sqrt{2} L_D} \frac{1 - e^{-\beta\psi_s} + (n_{p0}/p_{p0})(e^{\beta\psi_s} - 1)}{F(\beta\psi_s, n_{p0}/p_{p0})}$$

$$\rightarrow C_D(\psi_s \rightarrow 0) = C_D(\text{flat-band})$$

using Taylor expansion; $\psi_s = 0$ ↔ $V = 0$

$$C_D(\psi_s \rightarrow 0) = \frac{\varepsilon_s}{L_D} \quad (4.29)$$

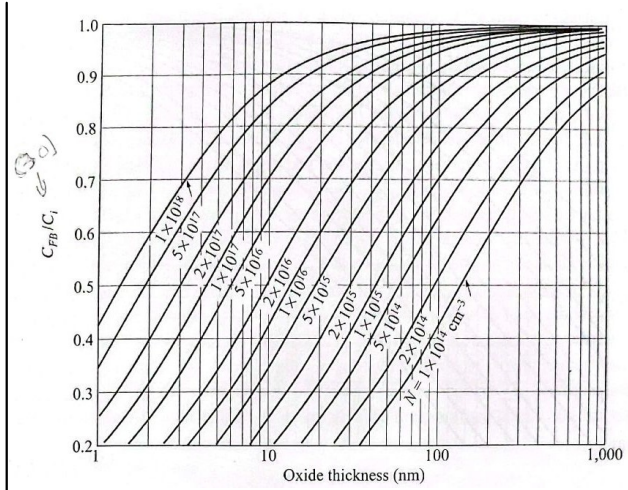
Total flat-band capacitance:

$$C(\text{flat-band}) = C(\psi_s \rightarrow 0)$$

$$= C_{FB} = \frac{C_i C_D(\text{flat-band})}{C_i + C_D(\text{flat-band})}$$

$$C_{FB} = \frac{\varepsilon_i \varepsilon_s}{\varepsilon_s d + \varepsilon_i \sqrt{\frac{kT\varepsilon_s}{N_A q^2}}} \quad (4.30)$$

$N_A$ ↔ $p_{p0}$



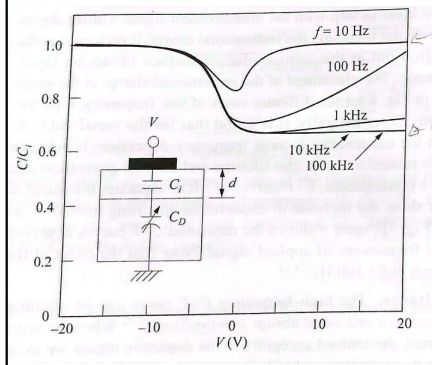




**Fig. 9** The frequency effect on $C$–$V$ curve of MIS capacitor with $N_A = 1.45\times10^{16}$ cm$^{-3}$ and $d = 200$ nm. Inset shows the series connection of the capacitors. The onset of low-frequency curves occurs at $f \leq 100$ Hz.

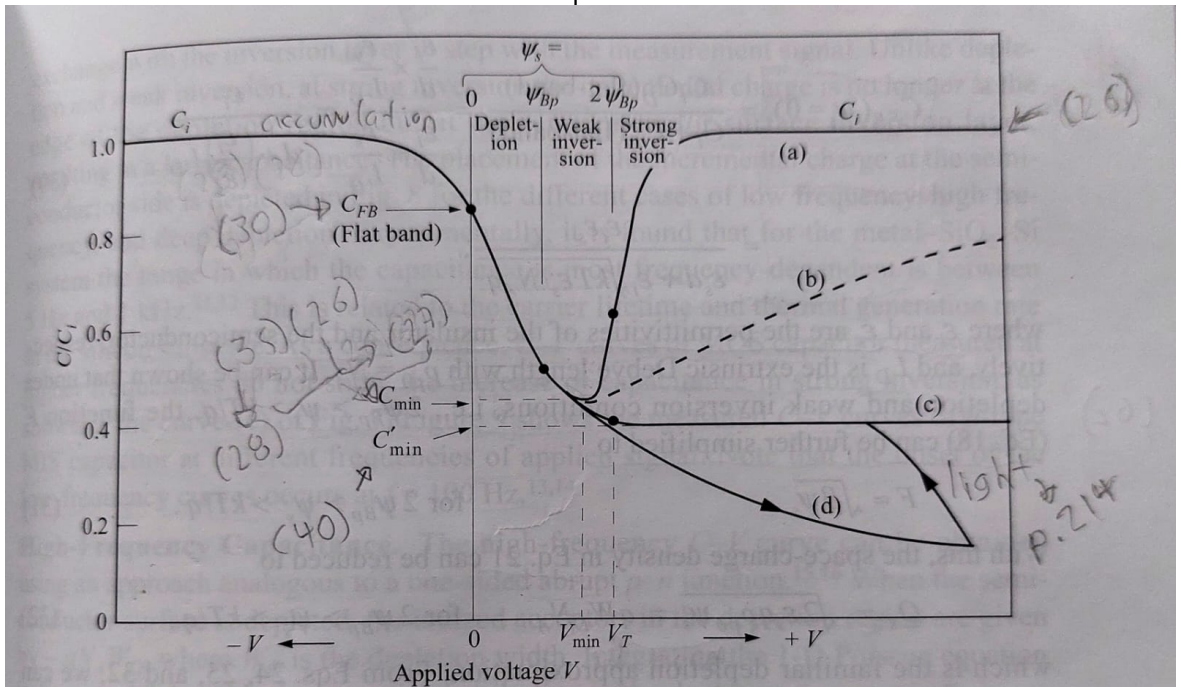



**Fig. 7** MIS *C–V* curves: (a) low frequency, (b) intermediate frequency, (c) high frequency, and (d) high frequency with fast sweep (deep depletion). Notably, the voltage is applied to the metal relative to the *p*-semiconductor and the flat-band voltage is assumed when $V = 0$.

*Under weak inversion:

$$\rightarrow \frac{n_i}{q} < \psi_s < 2\psi_{Bp}$$

$$\rightarrow F \approx \sqrt{\beta\psi_s} \qquad (4.31)$$

$$\rightarrow Q_S = \sqrt{2\varepsilon_s q p_{p_0} \psi_s} = q W_D N_A \qquad (4.32)$$

Depletion width: (4.33)

$$W_D = \sqrt{\frac{\varepsilon_s^2}{C_i^2} + \frac{2\varepsilon_s V}{q N_D}} - \frac{\varepsilon_s}{C_i}$$

$$\rightarrow C_D = \frac{\varepsilon_s}{W_D} = \sqrt{\frac{\varepsilon_s q N_A}{2\psi_s}} \qquad (4.34)$$

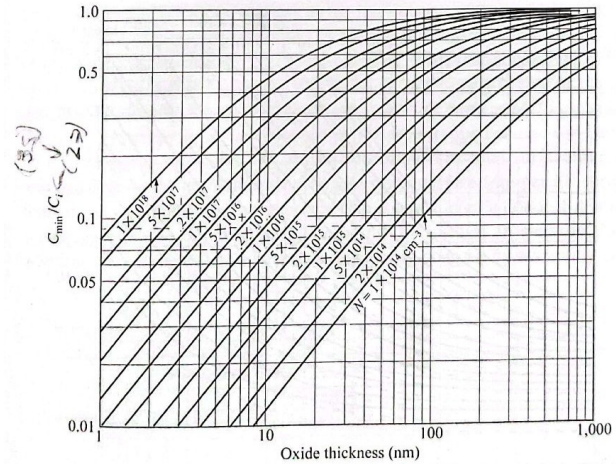
At low frequency: ~10 Hz (F. 4.g)

$C_{D\,min} \rightarrow C_{min}$

$$\hookrightarrow \left.\frac{dC_D}{d\psi_s}\right|_{C_{D\,min}} = 0 \rightarrow \psi_s = \psi_{s\,min} \downarrow V_{min}$$

(4.35)

$$\rightarrow \sqrt{\cosh(\beta\psi_{s_{min}} - \beta\psi_B)} = \frac{\sinh(\beta\psi_{s_{min}} - \beta\psi_B) - \sinh(-\beta\psi_B)}{\sqrt{\frac{N_A}{n_i}}\, F\left(\beta\psi_{s_{min}}, \frac{n_{p_0}}{p_{p_0}}\right)}$$



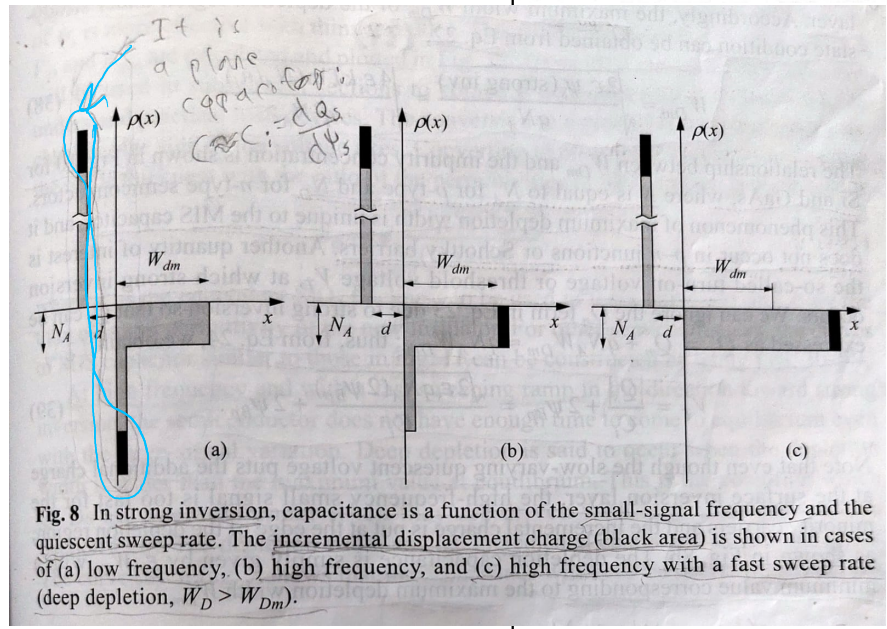




**Fig. 8** In strong inversion, capacitance is a function of the small-signal frequency and the quiescent sweep rate. The incremental displacement charge (black area) is shown in cases of (a) low frequency, (b) high frequency, and (c) high frequency with a fast sweep rate (deep depletion, $W_D > W_{Dm}$).

Potential distribution in the depletion region:

$$\psi_p(x) = \psi_s \left(1 - \frac{x}{W_D}\right)^2 \quad (4.36)$$

$$\psi_p(x) = \frac{q N_A W_D^2}{2\varepsilon_s} \left(1 - \frac{x}{W_D}\right)^2 \quad (4.37)$$

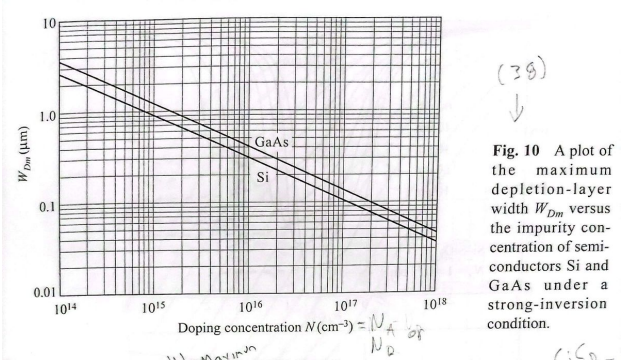
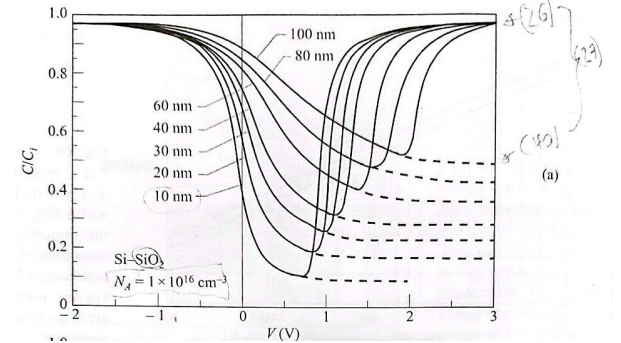



**Fig. 10** A plot of the maximum depletion-layer width $W_{Dm}$ versus the impurity concentration of semiconductors Si and GaAs under a strong-inversion condition.




Maximum $W_D$ under steady state condition at High-Freq.: (F ~ 10 mHz) and strong inversion: (F. 4.9)

$$\rightarrow \psi_s = 2\psi_B \leftarrow \psi_{Bp}$$

$$W_{Dm} \simeq \sqrt{\frac{2\varepsilon_s (2\psi_B)}{q N_A}} \quad (4.38)$$

$$= \sqrt{\frac{4\varepsilon_s kT \ln(N_A/n_i)}{q^2 N_A}}$$

$$\rightarrow Q_s = Q_n + q N_A W_{Dm} \simeq q N_A W_{Dm}$$

Threshold voltage:

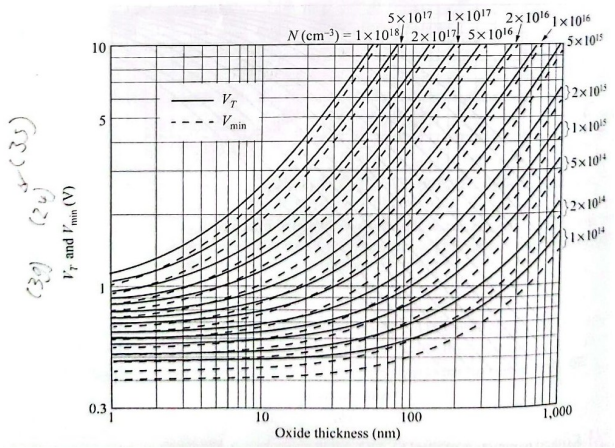
$$V_T = V_i + \psi_s$$

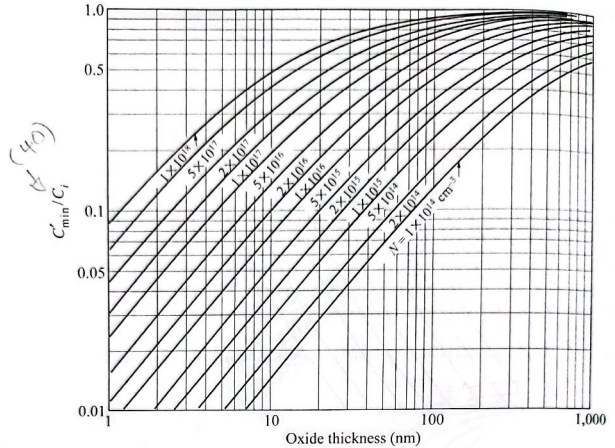
$$V_T = \frac{|Q_s|}{C_i} + 2\psi_B \quad (4.39)$$

$$= \frac{\sqrt{2\varepsilon_s q N_A (2\psi_B)}}{C_i} + 2\psi_{Bp}$$

$$\rightarrow C'_{min} = \frac{C_i C_D}{C_i + C_D} = \frac{\frac{\varepsilon_i}{d} \times \frac{\varepsilon_s}{W_{Dm}}}{\frac{\varepsilon_i}{d} + \frac{\varepsilon_s}{W_{Dm}}}$$

$$C'_{min} = \frac{\varepsilon_i \varepsilon_s}{\varepsilon_s d + \varepsilon_i W_{Dm}} \quad (4.40)$$




Counting MIS = M/SiO₂/Si curves scaling the insulator thickness to other insulators: (4.41)

$$d_{SiO_2} = d_i \frac{\varepsilon_i(SiO_2)}{\varepsilon_i(\text{insulator})}$$

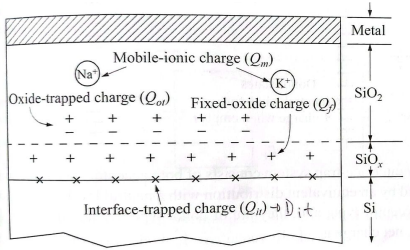



**Fig. 13** Terminology for charges associated with thermally oxidized Si. (After Ref. 18.)

$Q_{it}$ ← excess Si (trivalent Si)
Broken Si-H bonds
Excess oxygen
Impurities

$Q_f$ : Immobile under an $\mathcal{E}$.

$Q_{ot}$ ← X-ray radiation
Hot-electron injection
→ distributed in the oxide

$Q_m$ : mobile ions ← Temperature
stress conditions

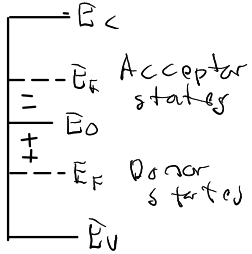
Interface traps can be high:

$$\frac{Q_{it}}{q} \sim 10^{15} \frac{atms}{cm^2}$$

Interface traps can be low:

$$\frac{Q_{it}}{q} \sim \frac{10^{10}\ atms}{cm^2} \leftarrow \frac{1\ int.\ trap}{10^5\ atms}$$

Interface traps: (4.43)




$$Q_{it} = -q\int_{E_0}^{E_F} D_{it}\,dE$$

$$Q_{it} = q\int_{E_F}^{E_0} D_{it}\,dE$$

$E_0$ : Neutral level

Interface trap density:

$$D_{it} = \frac{1}{q}\frac{dQ_{it}}{dE} \quad (4.44)$$

$$[D_{it}] = \frac{\#\ of\ traps}{cm^2\ eV}$$

$D_{it}$ experimentally:

$\Delta Q_{it} \leftarrow \Delta E_F \leftarrow \Delta \psi_s$
→ $D_{it}$

Distribution function for i.t. considered as donor: (4.42)

$$F_{SD}(E_t) = 1 - \frac{1}{1 + \frac{1}{g_D} e^{\frac{E_t - E_F}{kT}}}$$

$$= \frac{1}{1 + g_D\, e^{\frac{E_F - E_t}{kT}}}$$

$E_t$ : Energy of i.t.
ground state degeneracy for donor:

$$g_D = 4$$

Distribution function for i.t. considered as acceptor (4.42)

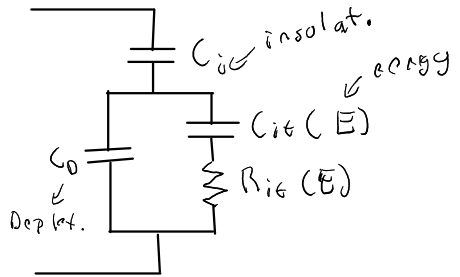
$$F_{SA}(E_t) = 1 - \frac{1}{1 + \frac{1}{g_A} e^{\frac{E_F - E_t}{kT}}}$$

$$= \frac{1}{1 + g_A e^{\frac{E_t - E_F}{kT}}}$$

$E_t$: Energy of i.t.

ground state degeneracy for acceptor: $g_A = 4$

Equivalent circuit including effects of the i.t.:



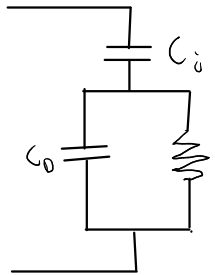

Lifetime of i.t.:

$$\tau_{it} = C_{it} R_{it}$$

Freq-dependant capacitance:

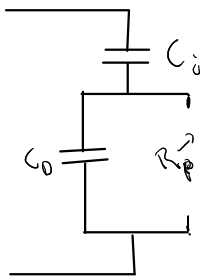
$$C_p \simeq C_D + \frac{C_{it}}{1 + \omega^2 \tau_{it}^2} \quad (4.45)$$

Freq.-dependant conductance:




$$G_p = \frac{C_{it} \omega^2 \tau_{it}}{1 + \omega^2 \tau_{it}^2} \quad (4.46)$$

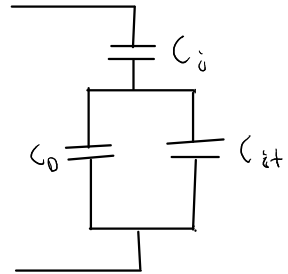
At high-freq:




Traps cannot respond to the fast signal

$$C_{HF} = \frac{C_i C_D}{C_{it} + C_D} \quad (4.48)$$

At low-freq: (4.47)




$$C_{LF} = \frac{C_i (C_D + C_{it})}{C_i + C_D + C_{it}}$$

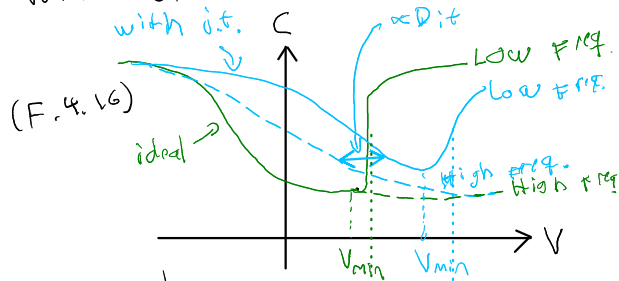
Conductance technique for measure Interface Traps

↓

Accurate results when MOS:

$$D_{it} \leq 10^{10} / cm \sim eV$$

With and without i.t.:




charge need Fill i.t.

↓

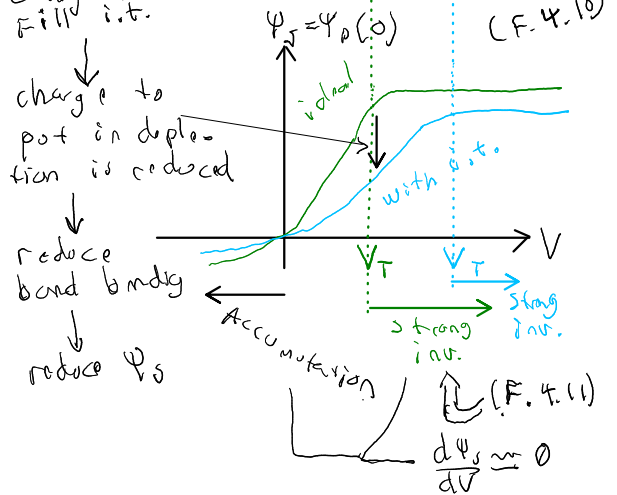
charge to pot in depletion is reduced

↓

reduce band bending

↓

reduce $\Psi_S$



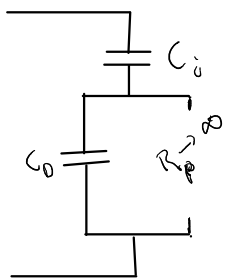
Capacitance due i.t.:

$$C_{it} = \frac{dQ_{it}}{d\psi_s} = \frac{(qD_{it}\,dE)}{(dE/q)} \leftarrow (4.44)$$

band bending (4.4)

$$C_{it} = q^2 D_{it} \quad (4.49)$$

At high-freq:

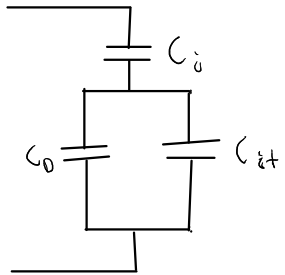



⇒ This is not $C_{it}$

but (4.28) $C_D \leftarrow \psi_s$

$C_D$ by change i.t.

Low-freq capacitance method:




in the semiconductor

portion of the voltage in the semiconductor: $\frac{d\psi_s}{dV}$

$$\left[\frac{d\psi_s}{dV}\right] = 1$$

Using C-V measure to obtain $\psi_s - V$: (4.50)

$$\frac{d\psi_s}{dV} = \frac{C_{LF}}{C_D + C_{it}} \leftarrow (4.47) = \frac{C_i}{C_i + C_D + C_{it}}$$

Without $C_D$: (4.52)

$$\frac{d\psi_s}{dV} = \frac{C_i}{C_i + C_D + C_{it}} = 1 - \frac{C_{LF}(V)}{C_i} \leftarrow \text{measure}$$

$$C_i = \frac{\epsilon_i}{d}$$

$$\psi_s(V_1) - \psi_s(V_2) = \int_{V_1}^{V_2} dV\left(1 - \frac{C_{LF}}{C_i}\right) \quad (4.53)$$

$$\frac{d\psi_s}{dV} \approx 0 \begin{cases} \text{starting accum.} \\ \text{or} \\ \text{starting strong} \\ \text{inversion} \end{cases} \quad \text{+ctte}$$

↳ (F.4.16)

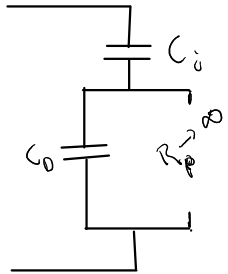
$$\rightarrow C_D(\psi_s) = \frac{dQ_s}{d\psi_s} = C_D(V) \leftarrow \psi_s(V) \quad (4.28)$$

$$= \frac{\epsilon_s}{\sqrt{2}L_D} \frac{1 - e^{-\beta\psi_s} + (n_{p0}/p_{p0})(e^{\beta\psi_s} - 1)}{F(\beta\psi_s, n_{p0}/p_{p0})}$$

Using $d\psi_s/dV$ to measure $D_{it}$: (4.51)

$$D_{it} = \frac{C_{it}}{q^2} = \frac{C_i}{q^2}\left[\left(\frac{d\psi_s}{dV}\right)^{-1} - 1\right] - \frac{C_0}{q^2}$$

High-freq capacitance method:




$$\frac{1}{C_{HF}} = \frac{1}{C_i} + \frac{1}{C_D} \quad (4.48)$$

$$C_D(V) = \frac{C_{HF}(V) C_i}{C_i - C_{HF}(V)} \to \frac{\epsilon_i}{d}$$

↳ measure

Obtain $\psi_s(V)$ from $C_D(V)$:

$$\rightarrow C_D(V) = \frac{dQ_s}{d\psi_s} \quad (4.28)$$

? This eq. is not in the book

$$= \frac{\epsilon_s}{\sqrt{2}L_D} \frac{1 - e^{-\beta\psi_s} + (n_{p0}/p_{p0})(e^{\beta\psi_s} - 1)}{F(\beta\psi_s, n_{p0}/p_{p0})}$$

Using $d\psi_s/dV$ to measure $D_{it}$: (4.51)

$$D_{it} = \frac{C_{it}}{q^2} = \frac{C_i}{q^2}\left[\left(\frac{d\psi_s}{dV}\right)^{-1} - 1\right] - \frac{C_0}{q^2}$$

High-Low-Freq. capacitance method:

$$C_{it} = \left(\frac{1}{C_{LF}} - \frac{1}{C_i}\right)^{-1} - C_D \qquad (4.54)$$

$$\hookrightarrow \left(\frac{1}{C_{HF}} - \frac{1}{C_i}\right)^{-1}$$

Capacitance gap: → Measure

$$\Delta C = C_{LF} - C_{HF}$$

$$\rightarrow D_{it} = \frac{C_{it}}{q^2} = \qquad (4.55)$$

$$\frac{\Delta C}{q^2}\left(1 - \frac{C_{HF} + \Delta C}{C_i}\right)^{-1}\left(1 - \frac{C_{HF}}{C_i}\right)^{-1}$$